

REVISITING RDMA RELIABILITY FOR LOSSY FABRICS

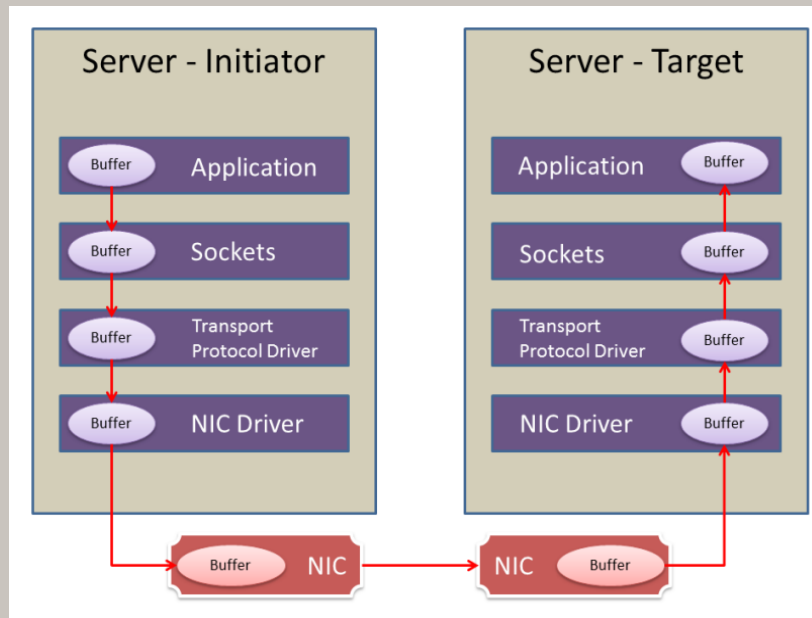
Aditya Sujith Gudimetla

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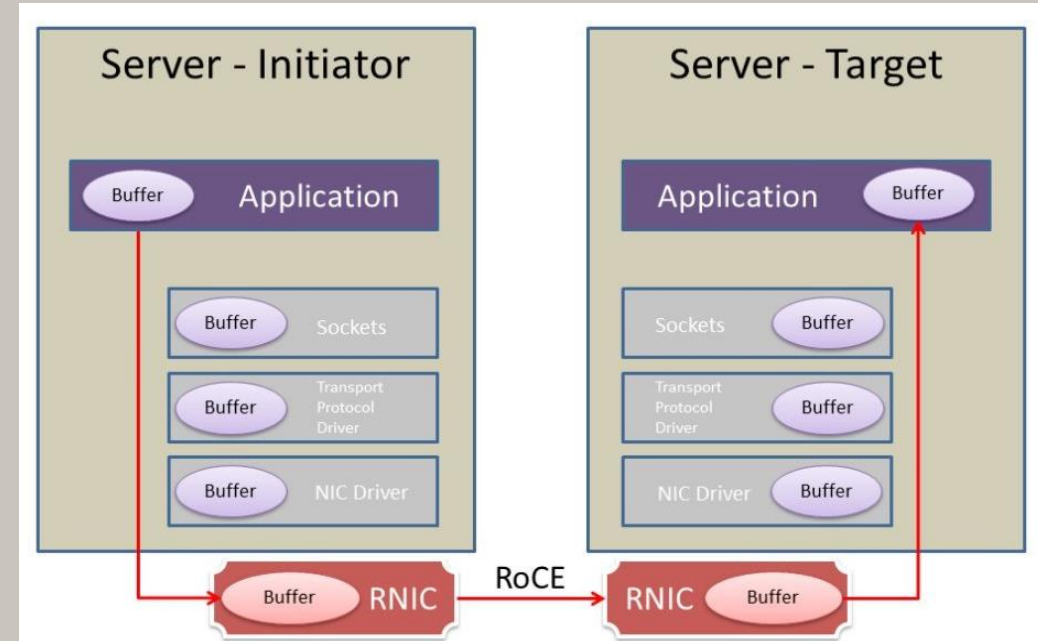
- What's RDMA ?
- Why lossy ?
- How is lossy dealt with ?
- Why lossy bad ?
- How to make lossy good ? DCP !
- How does DCP work ?
- Evaluation
- Questions

WHAT'S RDMA ?

- Before that, what's TCP ?



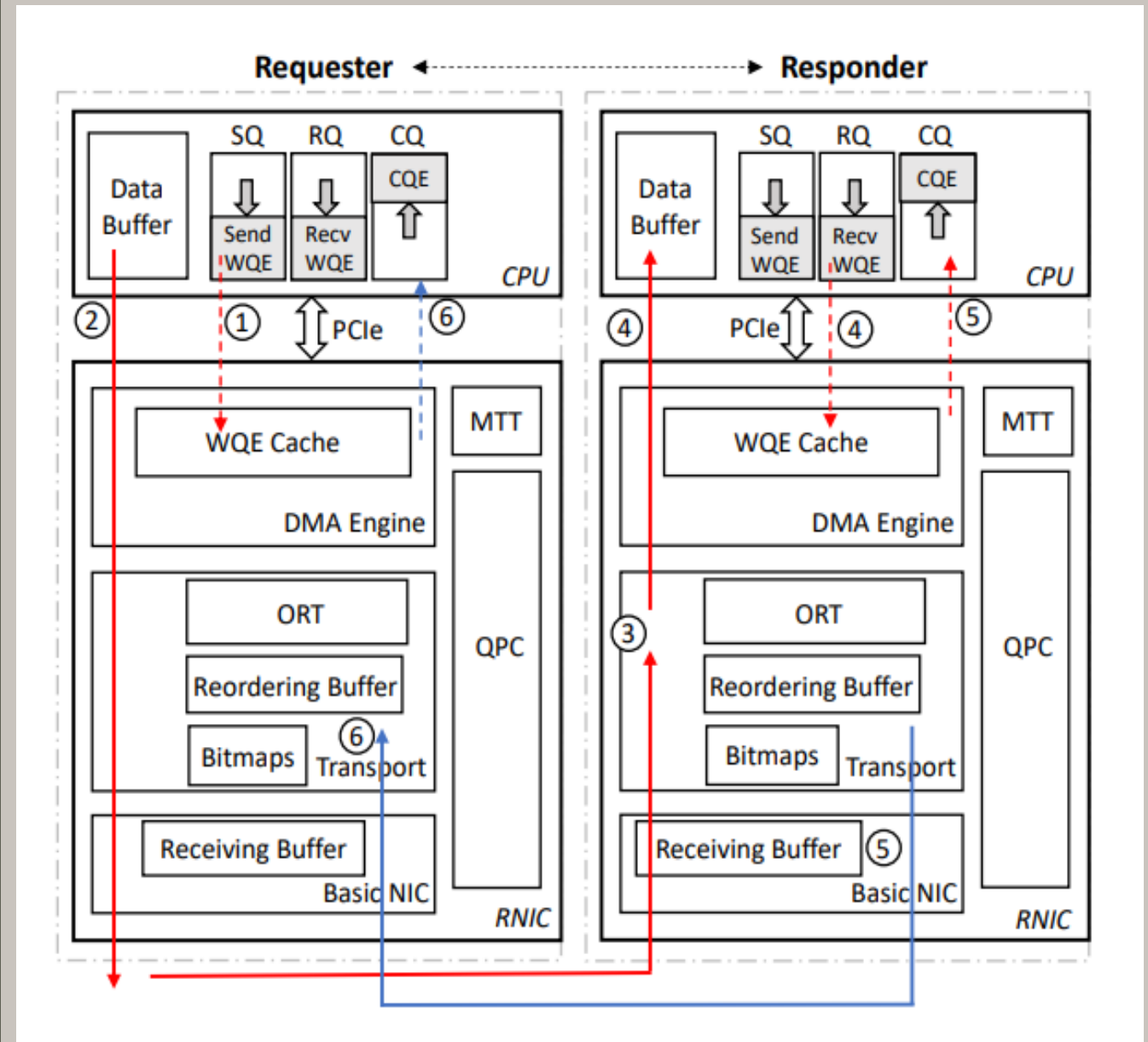
- Now, what's RDMA ?
 - Kernel Bypass
 - Zero Copy
 - Computation Offload



**RoCE : RDMA on Converged Ethernet*

WHAT'S RDMA REALLY ?

- Simply...



HOW'S IT WORK ?

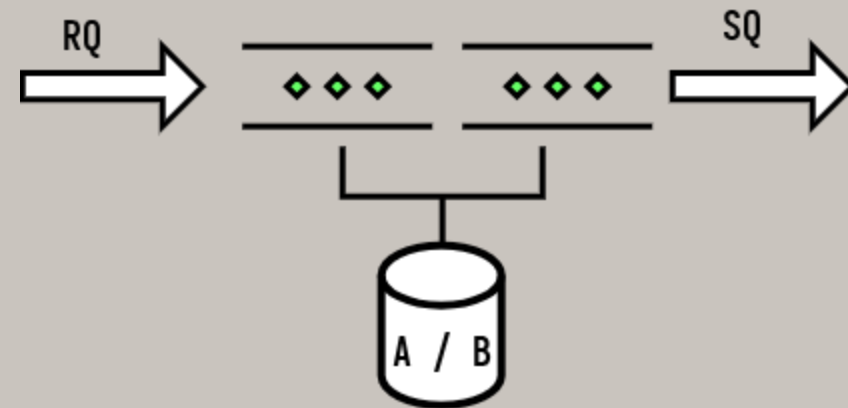
Part 1 : Initial

There's some data to send between 2 devices (hosts / servers / computers).



Each device has a Queue Pair (Queue Pair) and a QP Number (QPN).

A queue pair is a Send Queue (SQ) and a Receive Queue (RQ)

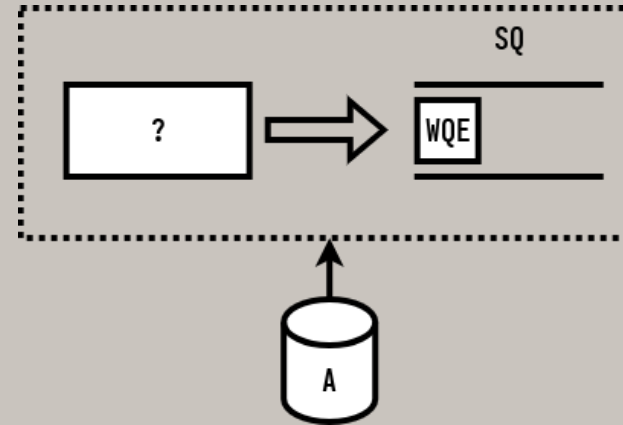


HOW'S IT WORK ?

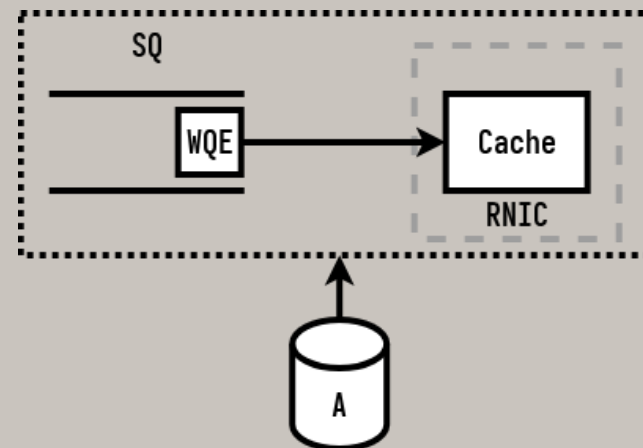
Part 2.1 : Sending

The SQ receives a Work Queue Element (WQE)

- Doorbell mechanism : The device notifies the RNIC that there is an WQE waiting in the SQ



The RNIC fetches the WQE and copies it to the WQE Cache

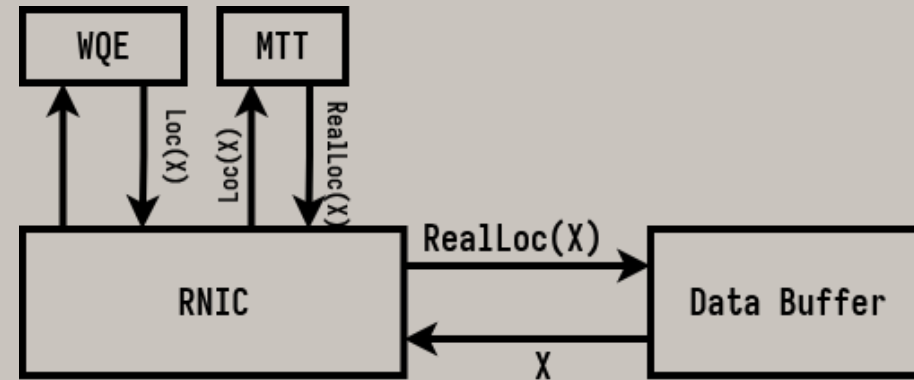


HOW'S IT WORK ?

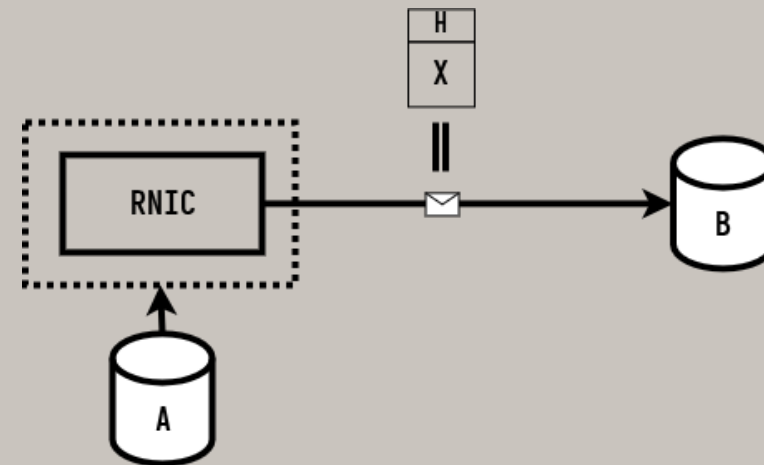
Part 2.2 : Sending

RNIC looks up the real location of X in memory from the MTT

- For retransmission efficiency, metadata stored in Outstanding Request Table (ORT)



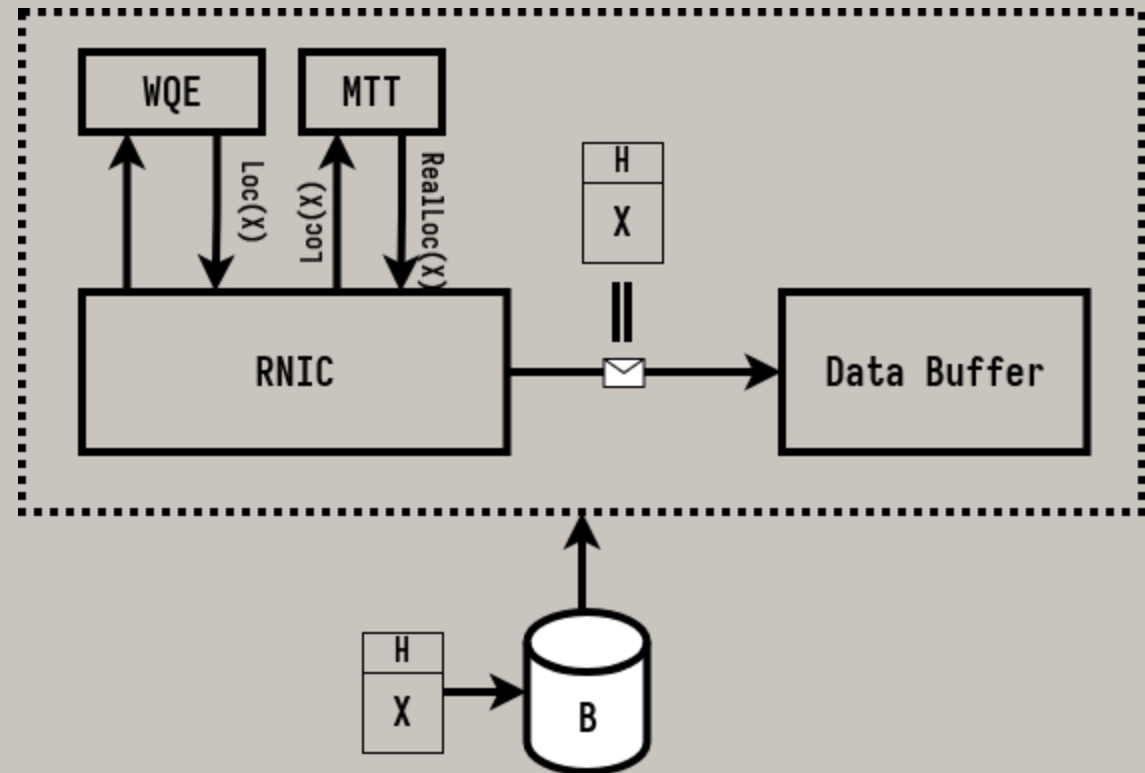
RNIC attaches headers then sends data to receiver



HOW'S IT WORK ?

Part 3 : Receiving

1. The packet arrives at B
2. RNIC checks MTT table for the correct position to write to.
3. Data is written to the calculated position



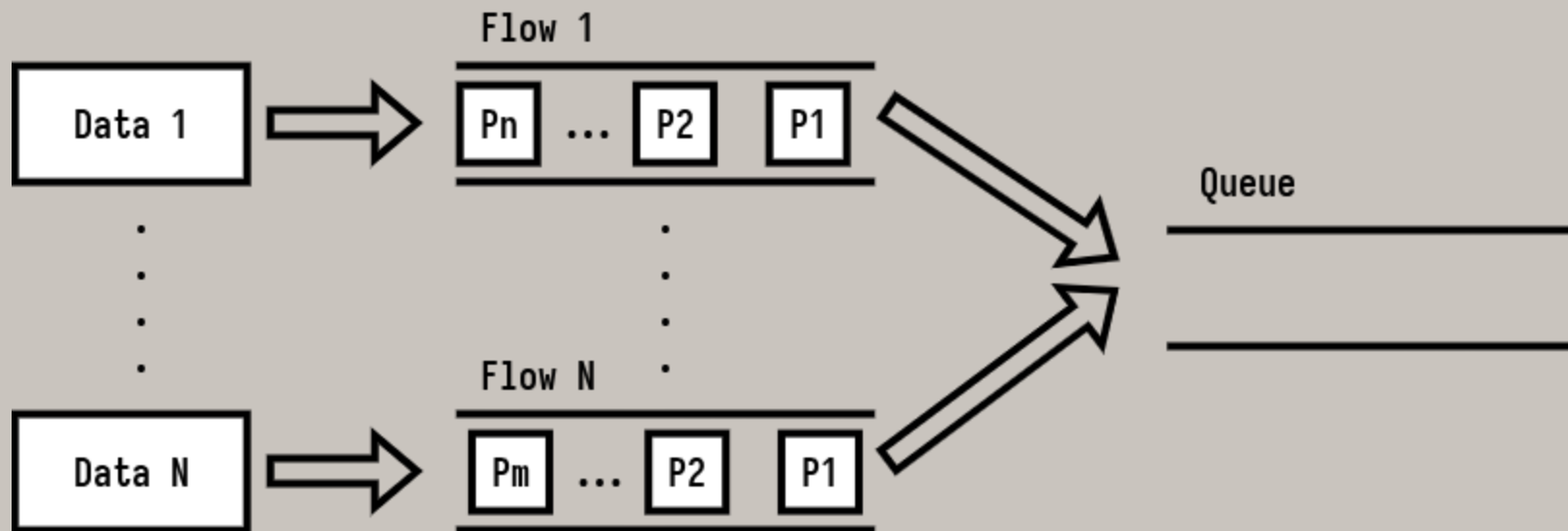
THE LIES

- Things not mentioned
 - Reordering Table
 - Outstanding Request Table
 - Bitmap
 - Ack Signals
 - Completion Notifications

WHY LOSSY ?

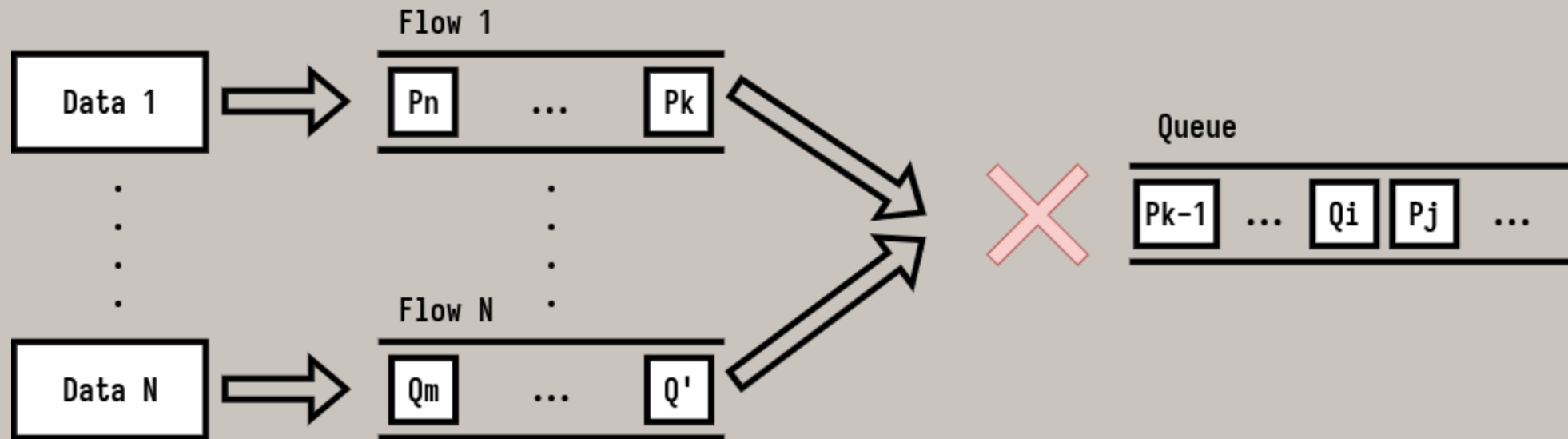
Part 1

- I. Fabric : The collection of switches and wires (a.k.a network)
- II. Packets have flow i.e sequential
- III. Switches have queues
- IV. Lossless prevents packet drops



- Priority-based Flow Control (PFC)
- When switch queue is full*, stop
- Different Flows can have different priorities
- Prevents packets being dropped due to congestion at queues

- Motivation for PFC
 - Packets need to arrive in order (GBN recovery is inefficient)
 - Stop taking more packets that will drop.
 - Wait till queue drains to take more



*strictly speaking, predefined threshold exceeded

- Drawbacks of PFC
 - HOL Blocking
 - Cascading Bottlenecks (congestion spreading)
 - Distance Deployment Issues (due to in flight packets + storage)
- Eliminating PFC a good idea
- So lossy good

HOW IS LOSSY DEALT WITH?

- Selective Repeat
- Better than GBN
- Does not need lossless fabric
- Needs only the packet needed to be sent
- Extra machinery needed (Goto 9)

WHY LOSSY (STILL) BAD ?

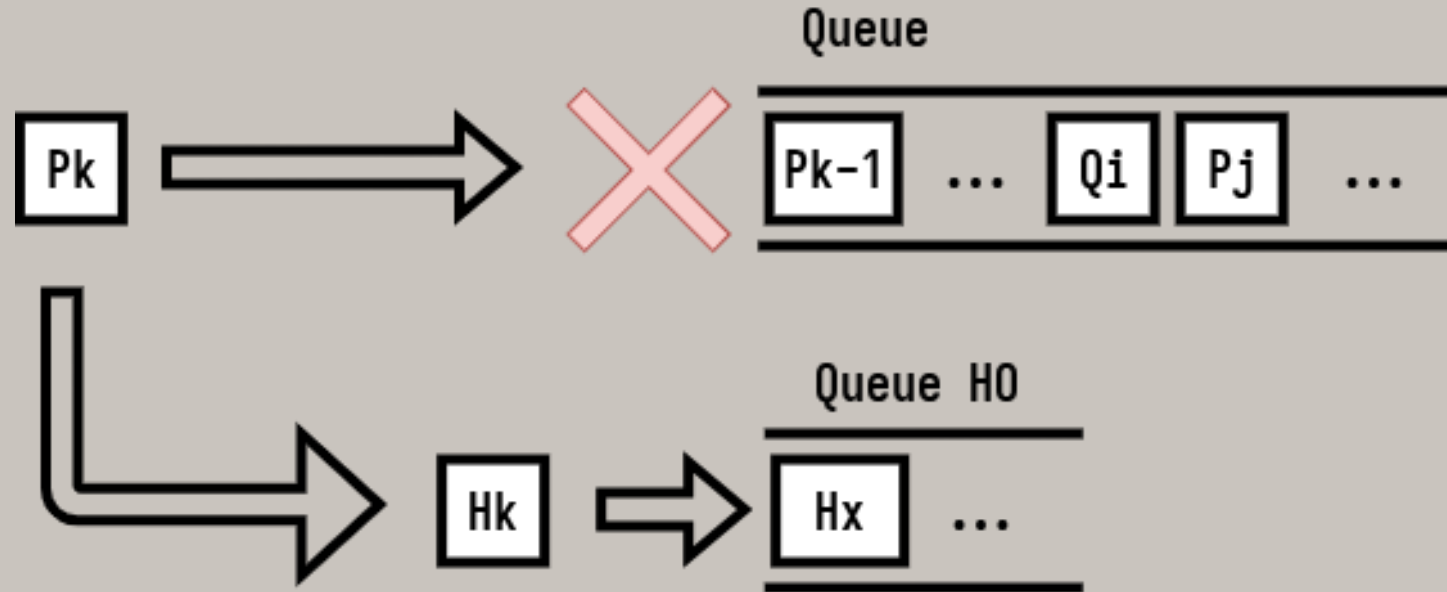
- SOTA is RNIC-SR
- RNIC-SR
 - 000 is considered packet loss and triggers retransmission
 - So is incompatible with Packet Level LB
 - Cannot replace ECMP-LB
 - Hash collisions cause throughput degradation
 - RTO caused by
 - Loss of last packet
 - Loss of Retransmitted packet

HOW TO MAKE LOSSY GOOD? DCP!

- What's DCP ?
 - Transport architecture
 - DCP-Switches
 - DCP-RNICs
- DCP why ?
 - Supports 000
 - No PFC
 - No RTO (mostly)

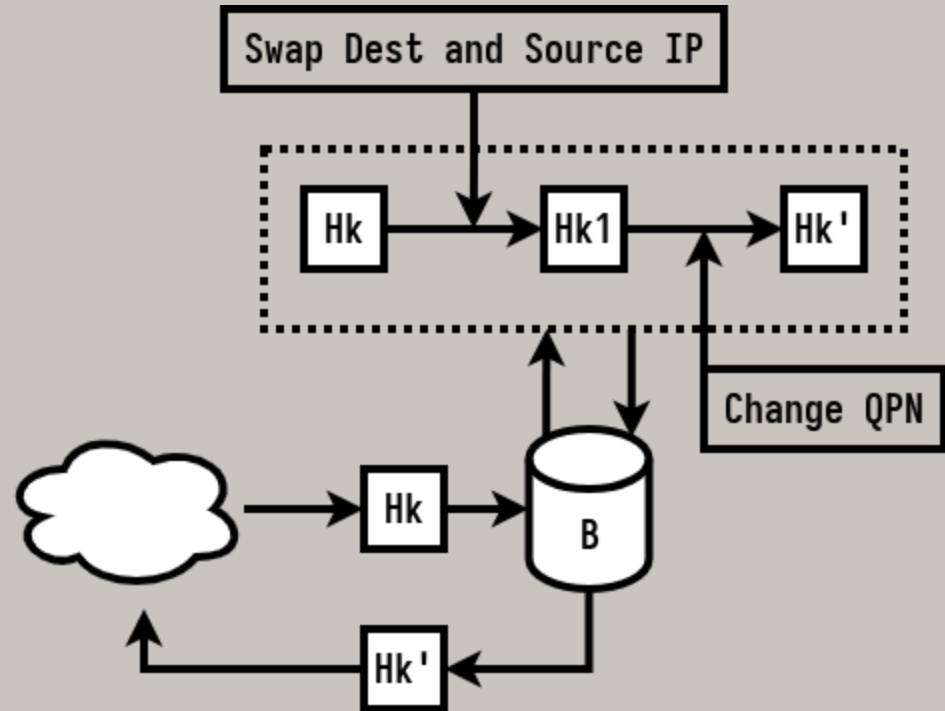
HOW DOES DCP WORK

- How does DCP work ?
 - If queue is full*, trim payload and send to header only queue.

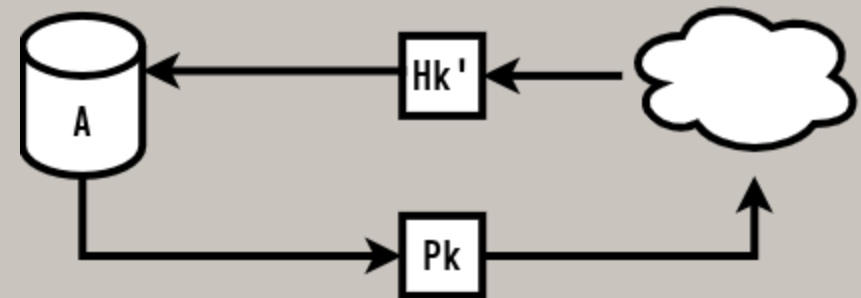


*strictly speaking, predefined threshold exceeded

- Receiver sees H0 packet
 - Swaps the Source and Destination IP
 - Replace the Dest QPN with Source QPN
 - Sends back

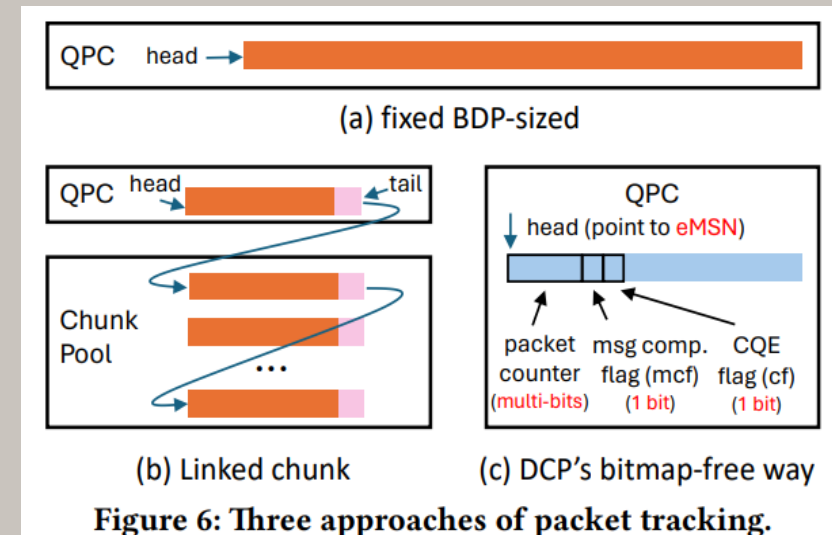
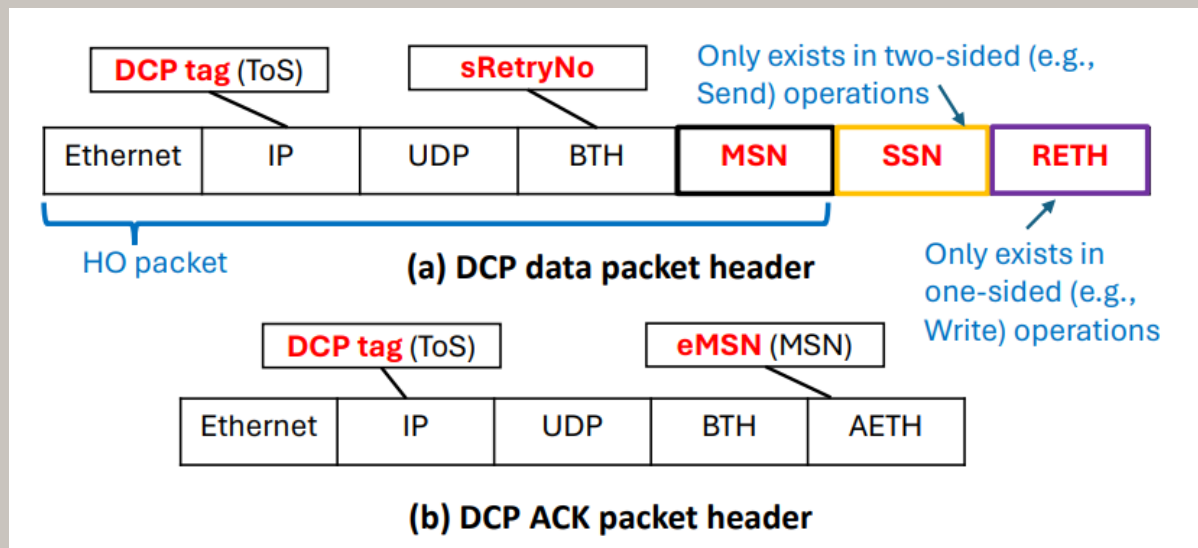


- Sender sees H0 packet
- Retransmits the original packet



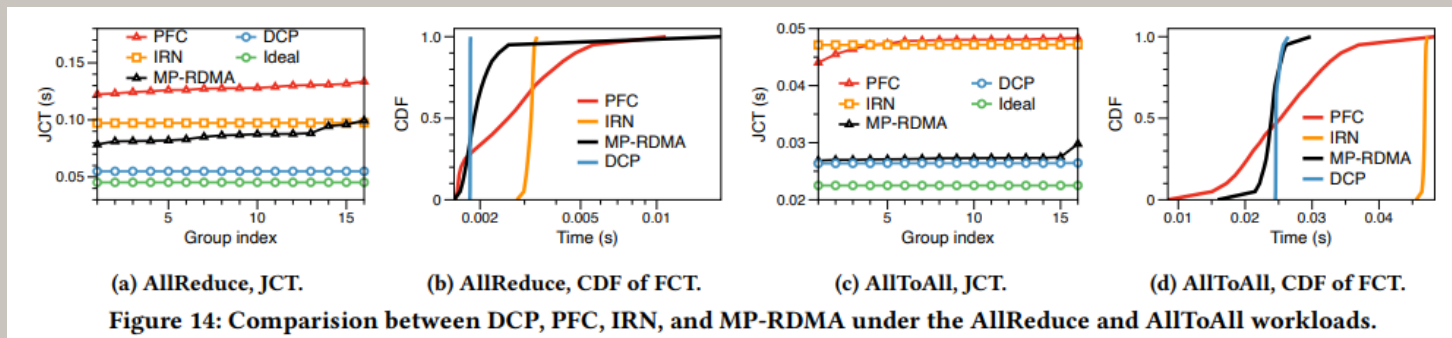
MORE DCP

- Packets only arrive once
 - Simply count packets, no need for bitmap
- Packets can arrive OOO
 - Are placed directly in memory
 - Possible due to header extensions



EVALUATION : EVEN MORE DCP

- AI Workload
 - AllReduce and AllToAll benchmarked
 - 300mb each operation
 - 16 servers x 16 groups = 256 machines
 - DCP
 - AllReduce: 38%, 44%, 61% lower average job completion time than MP-RDMA, IRN, PFC, respectively
 - AllToAll: 5%, 45%, 46% lower average job completion time



- Tested Configurations
 - PFC (RNIC-GBN)
 - IRN : Improved RoCE NIC
 - MP-RDMA : Multipath RDMA
 - DCP

QUESTIONS (I HAVE)

- What are the scales we're dealing with ?
 - What are the standard assumptions / calculations of the field (or RDMA) wrt practicality of algorithms ?
 - What are practical numbers / impractical numbers in terms of memory and time ?
 - Any standard assumptions of the machines themselves that can be made ? What are the limitations of the hardware ?
- Why is hardware offload an advantage of RDMA if there's way too many limitations on the size and computational power of the RNIC ?
- When the RNIC fetches the data from the data buffer of the host memory, does it only get how much it can fit into memory and leave the rest ?
 - If no, does that mean the host allocates chunks that would fit on the RNIC ?
 - If yes, how does the RNIC keep track of the chunks ?
 - Eg : What happens when I want to transfer a 5gb file ?
- Can we inspect the code ?
- Why not a run a complete training run instead of simulations specifically for the AI test ?

REFERENCES

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